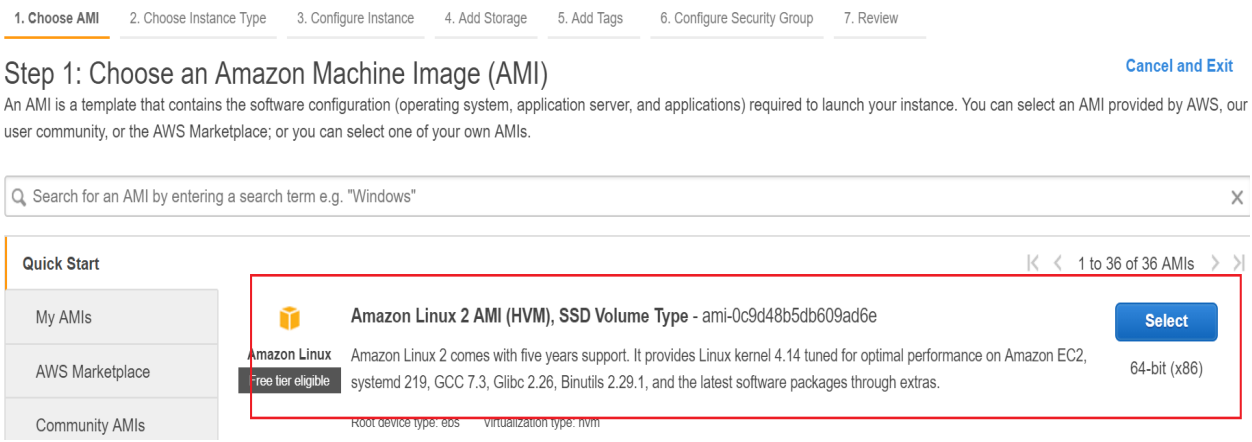
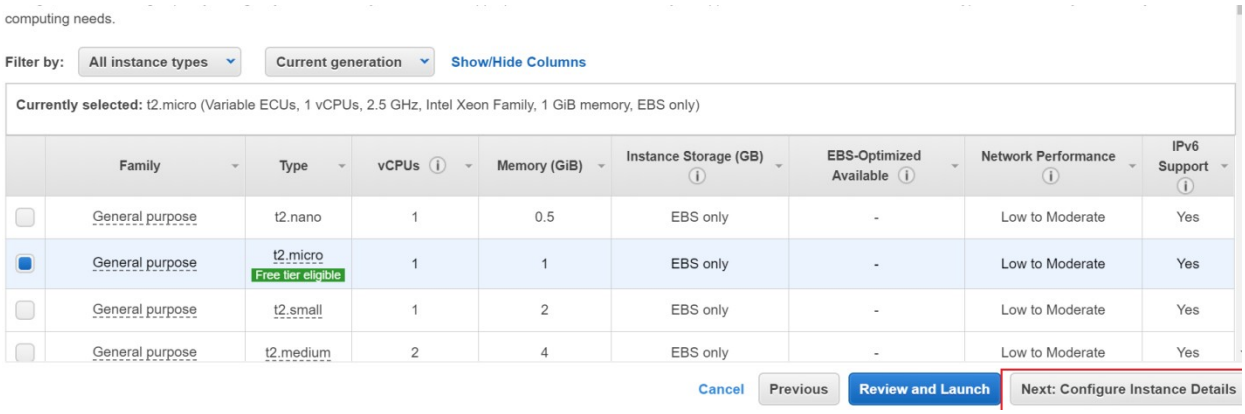
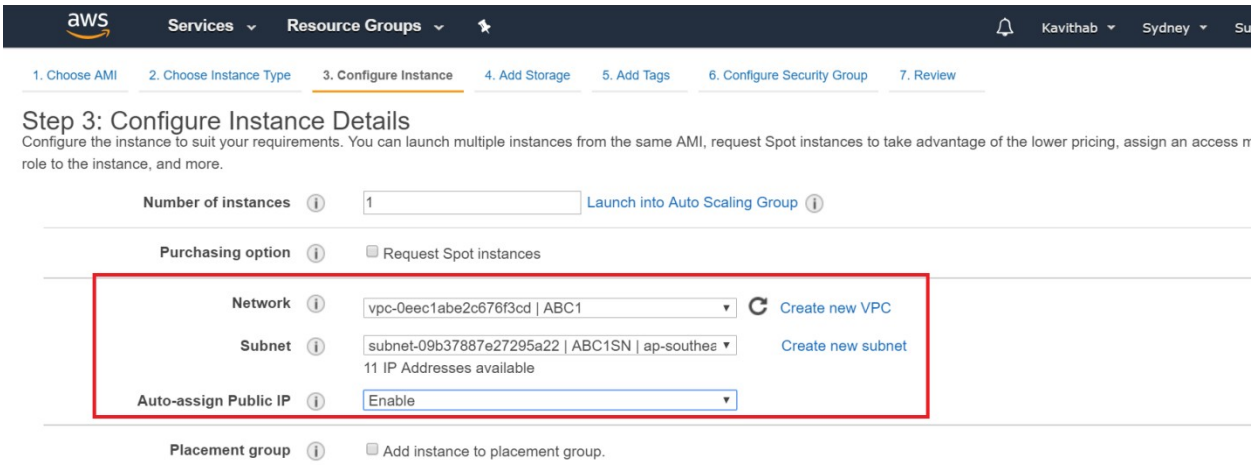
# Cloud Watch Steup

## Step1: -



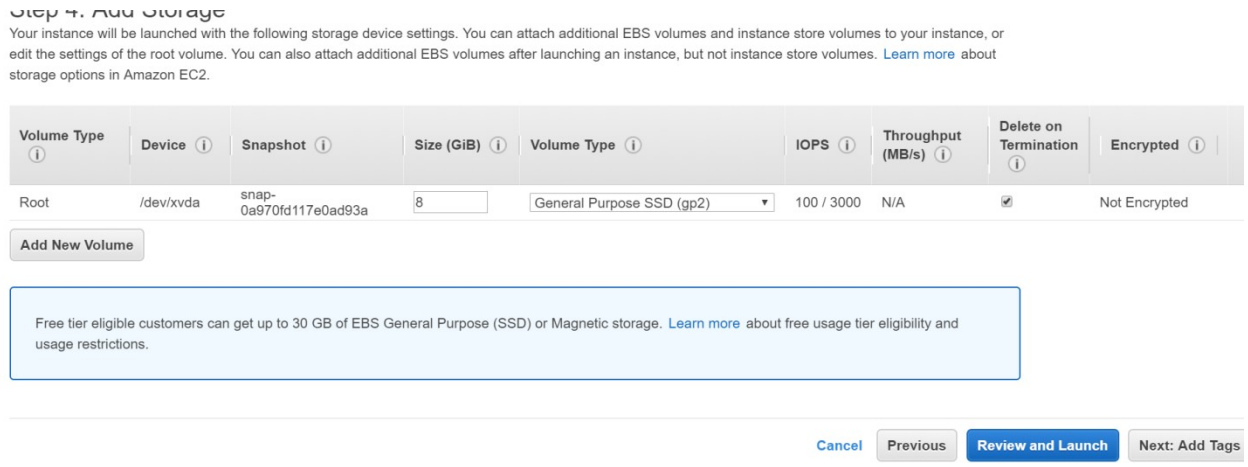

**Click on Configure instance details**



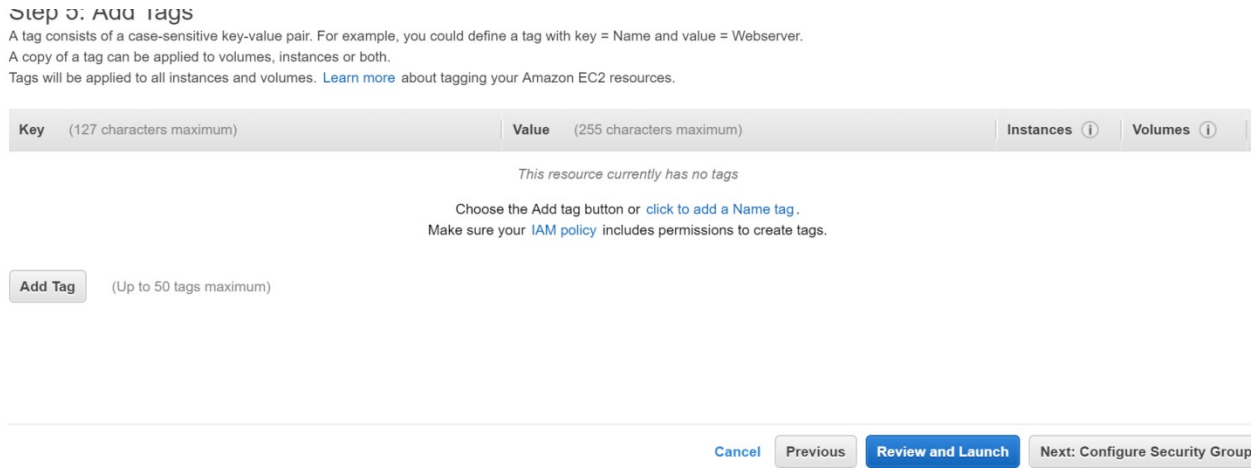

**Enter the details of Network (VPC)**

**Subnet (auto selection) and enable the Public IP as in the above screen shot.**

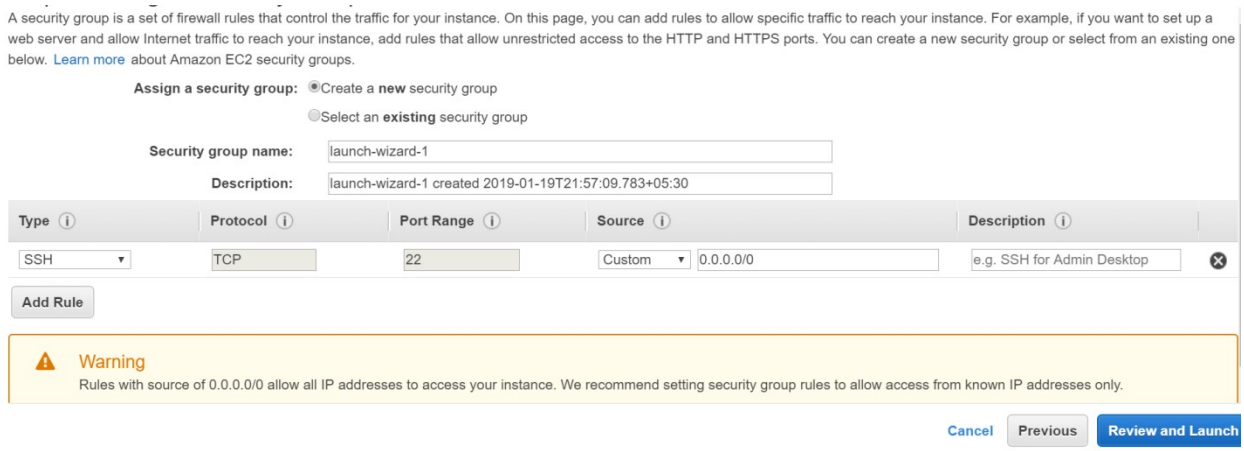
**Then click on add storage**



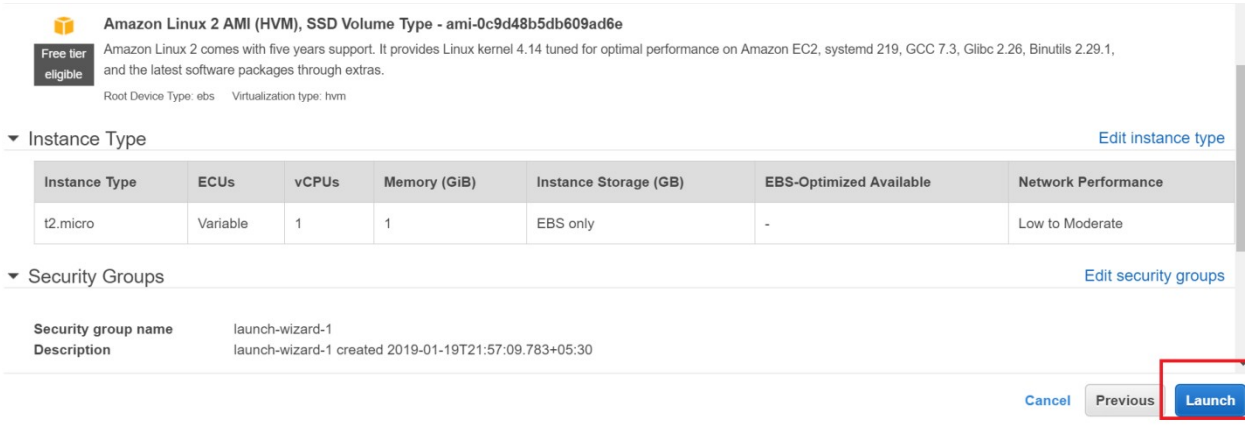

**Click on Add tag (nothing to be enter in the above screen)**



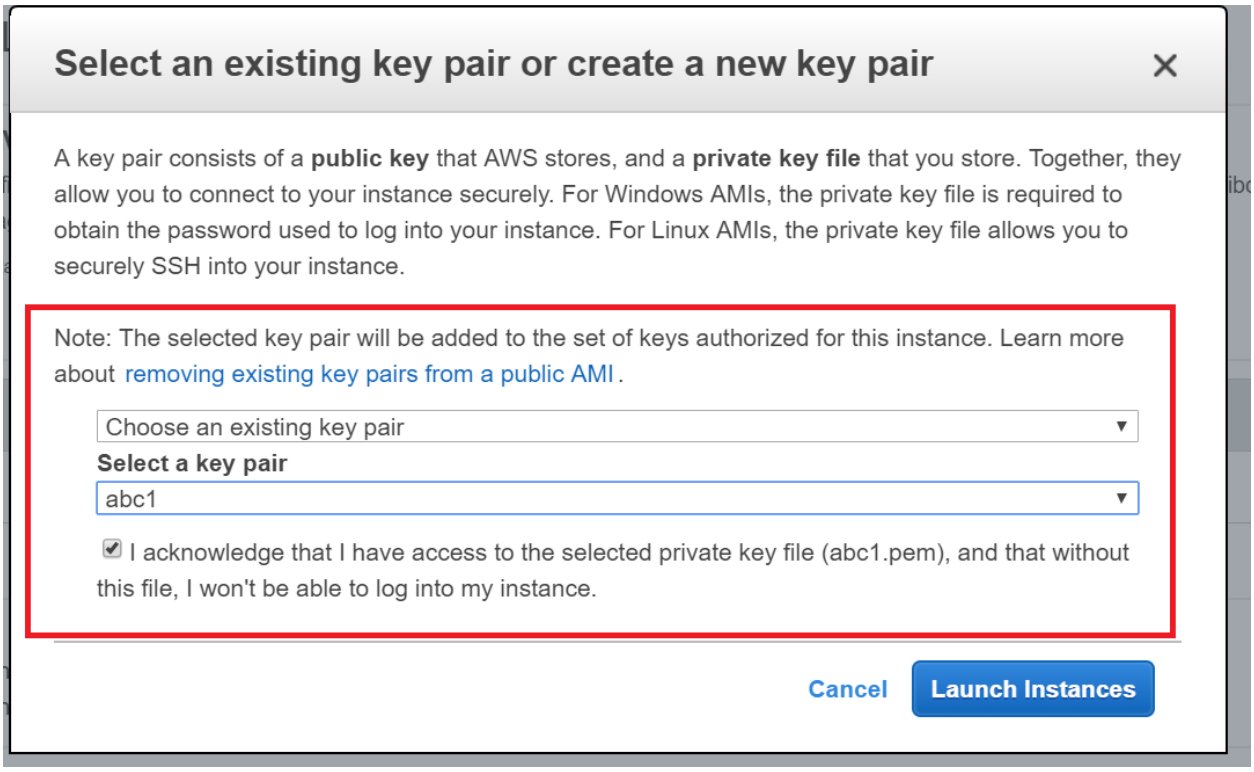
## Click on Configure Security Group




## Click on Review and launch

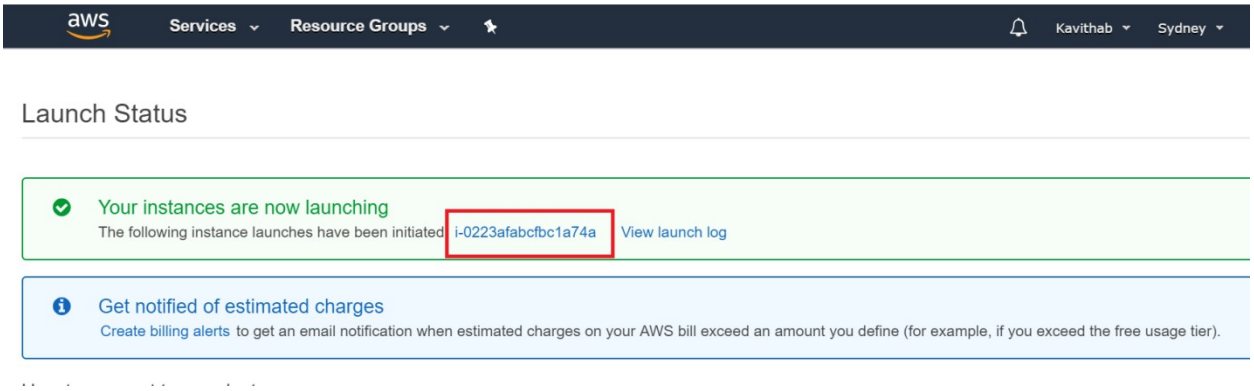



## Click on Lanuch

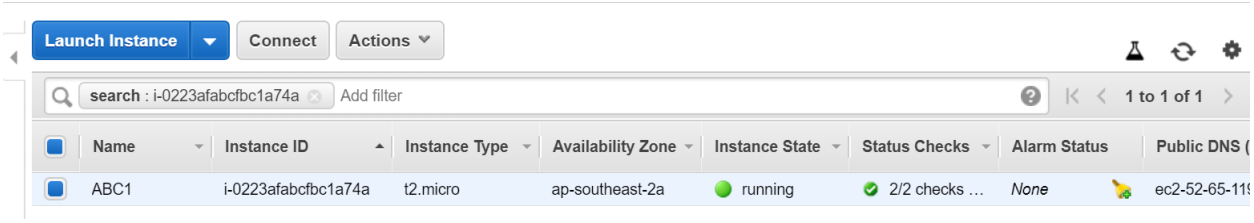



**Choose the existing pair key or create the new one**

**Then click on Launch Instances**

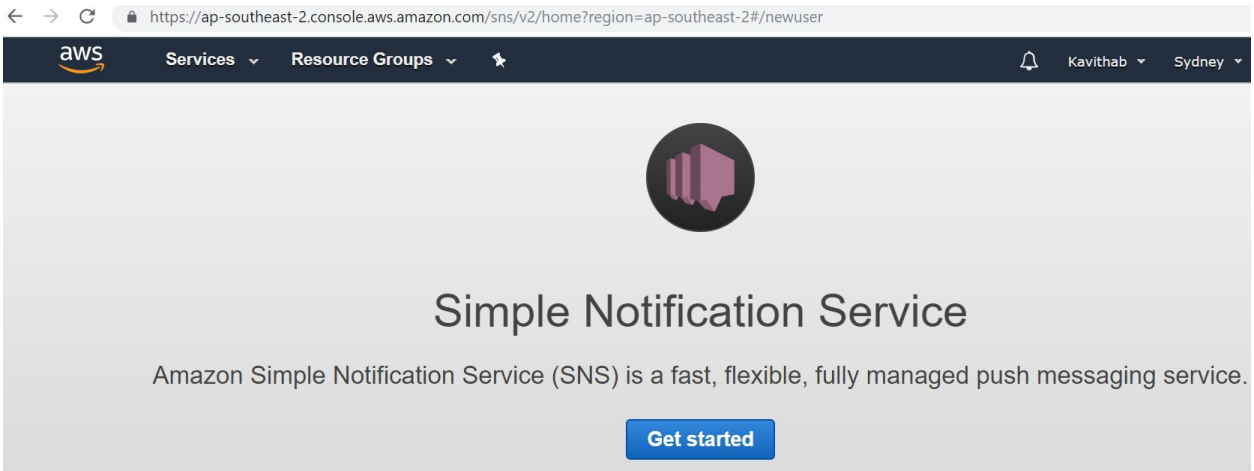



**Click on the id which is marked in the above screen**

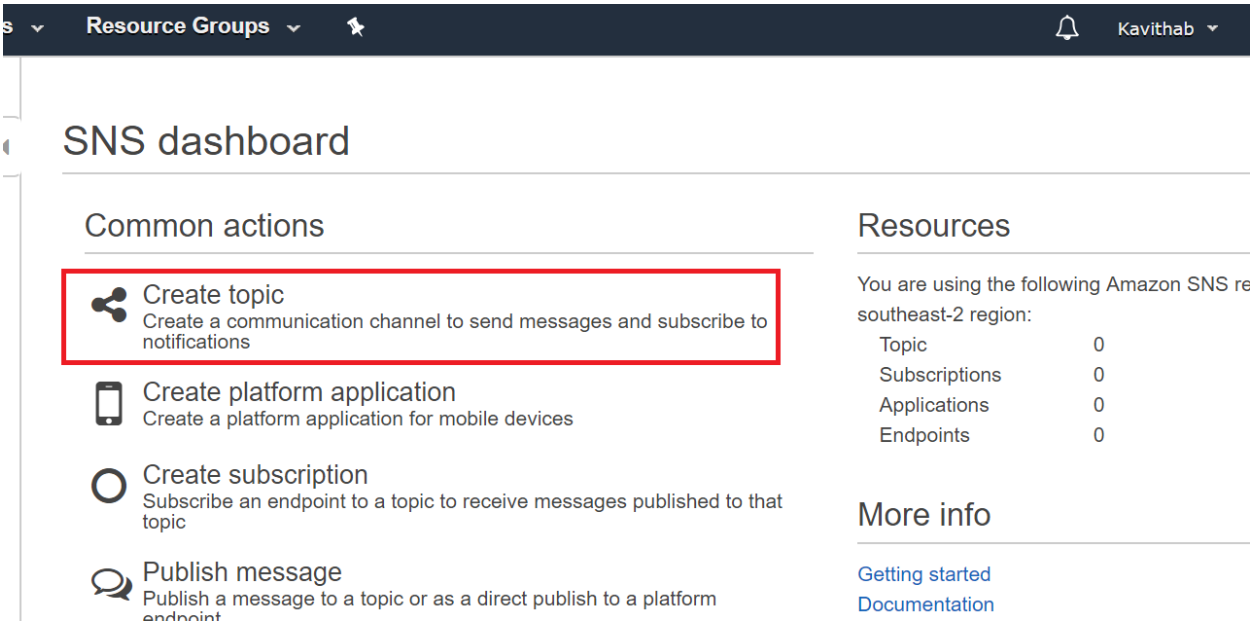



**Your instance is Launched now.**

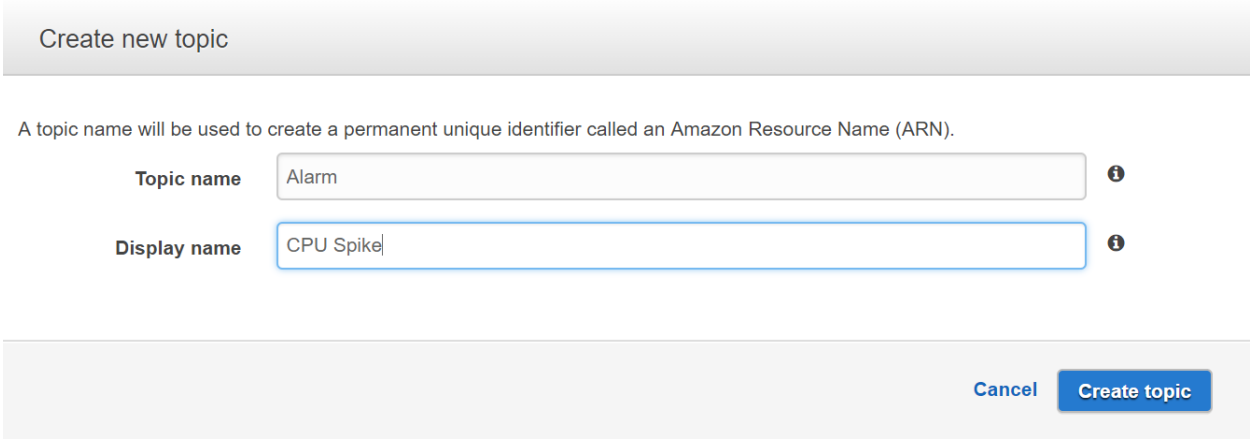
**From the services select Simple Notification Services**




**Click on Get Started**




**Click on Create Topic**



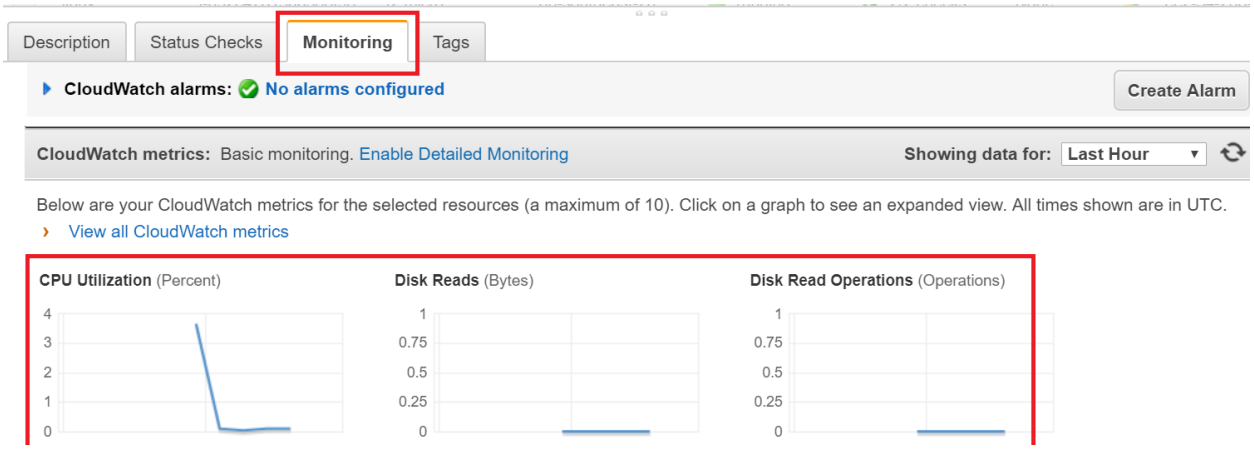

**Enter the Topic name**

**Display name (Incident topic ) In the above screen shot we are generating alarm for CPU Utilization, set display name as CPU Spike**

**Then click on Create Topic**

**Go to Instance**

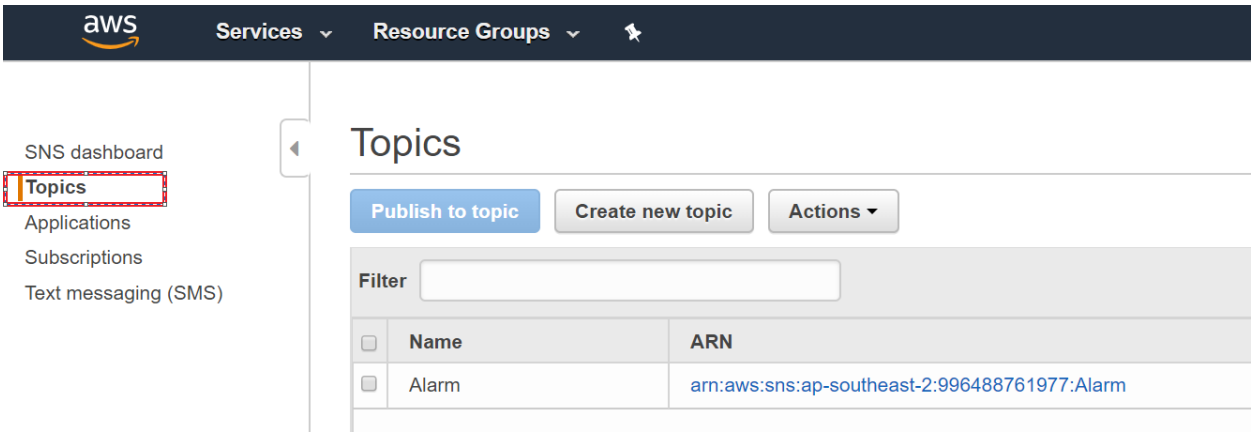
**Click on Monitoring**

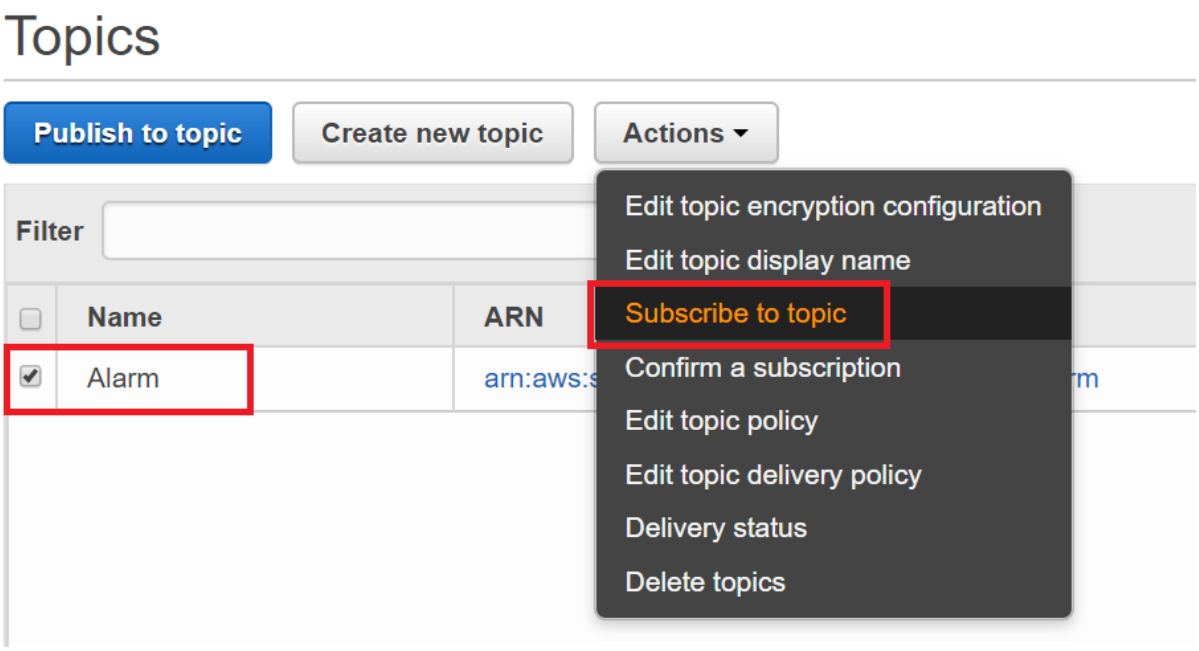



**We can view the Cpu utilization details**

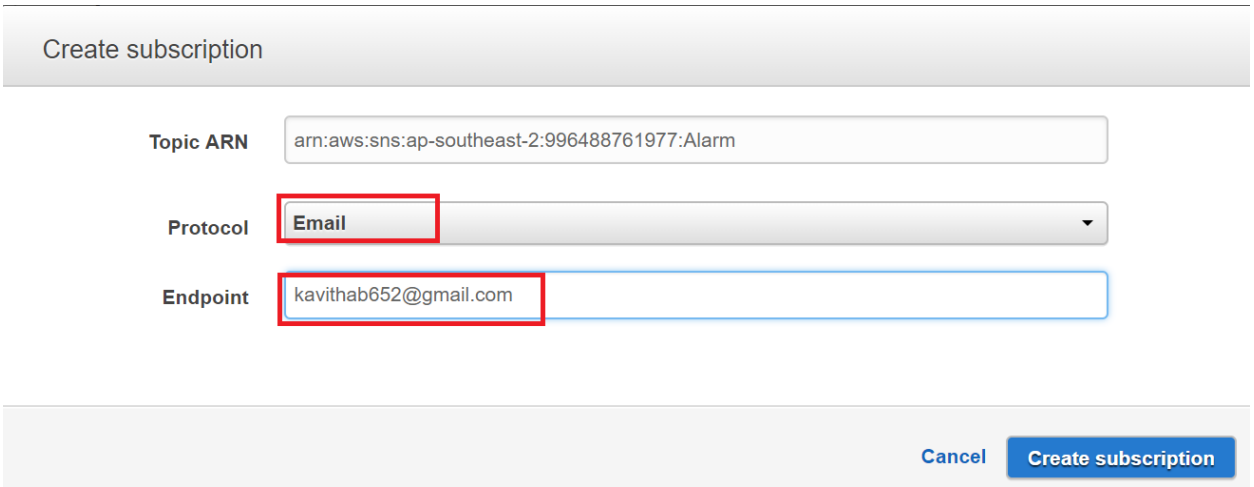
**Then go to SNS (simple notification services)**

**Click on Topics**




**Select your topic  and from Actions select Subscribe topic**

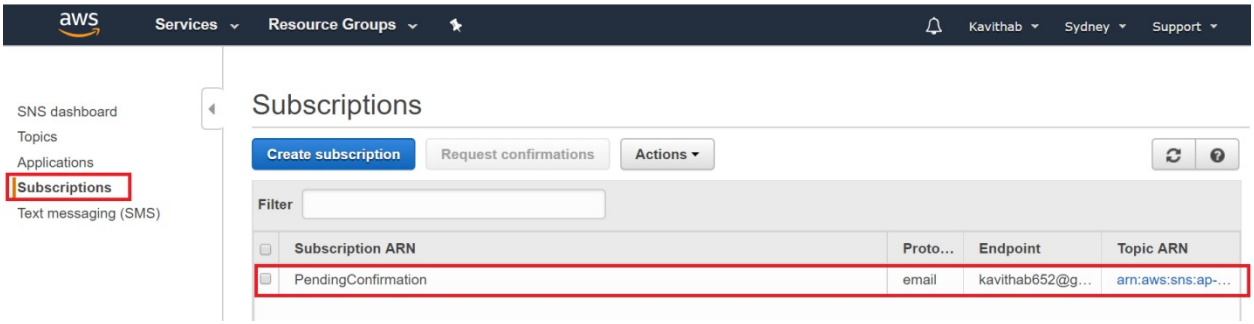



**Protocol ---Select your Services as Email**

**Endpoint --- Mail id**

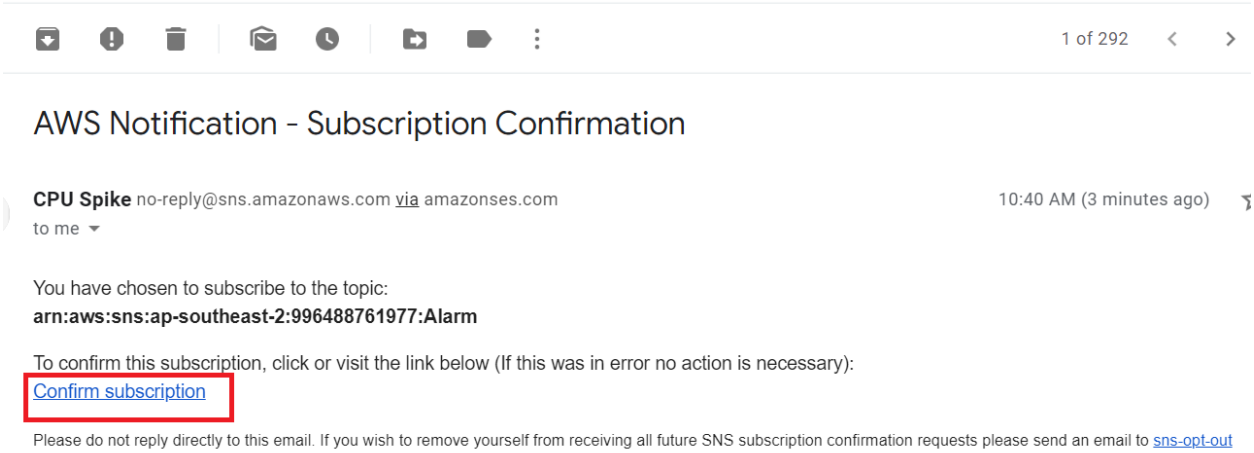
**Click on Create Subscription**

**Click on subscription**

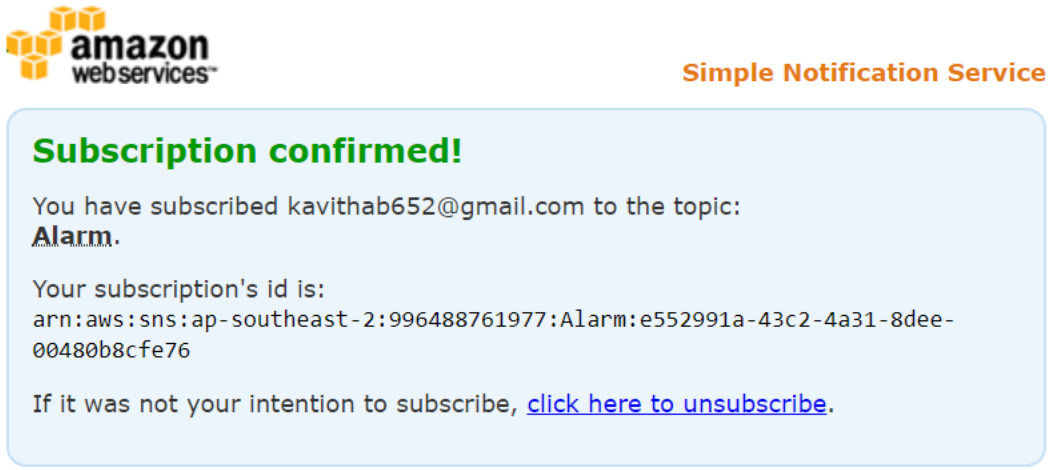



**You will get a mail for confirmation for the alerts.**

**Go to your mail box**




**Open the notification mail and click on Confirm Subscription**

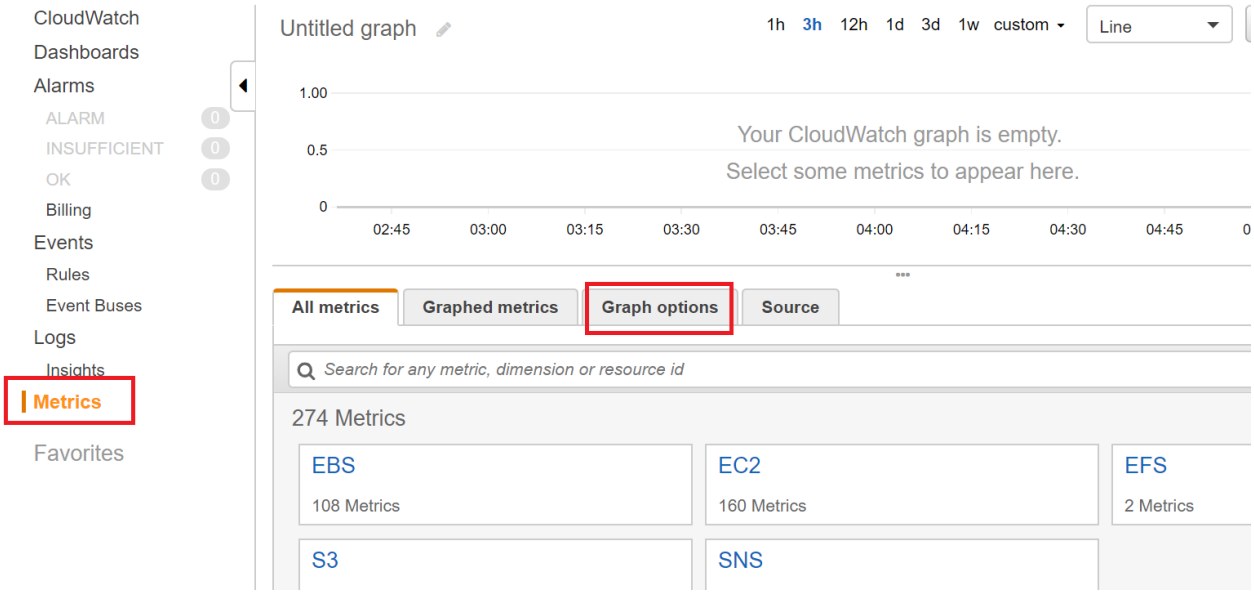



**Then go to SNS and click on Refresh button**

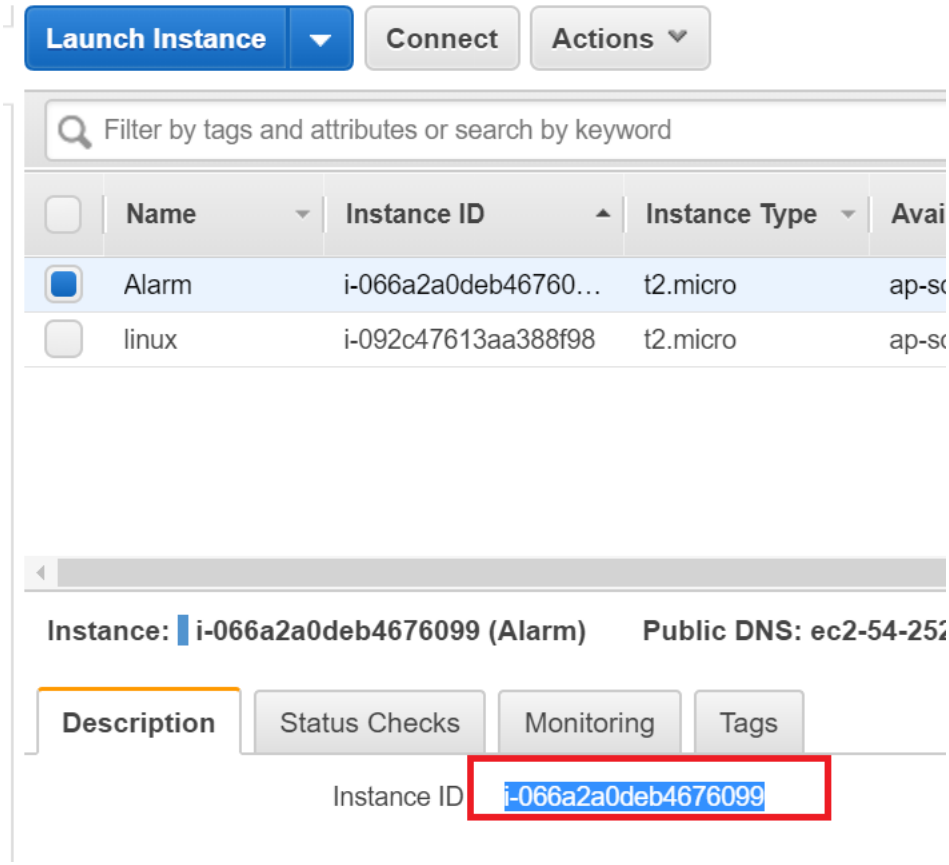
**Then go to services**

**Select Cloud Watch**

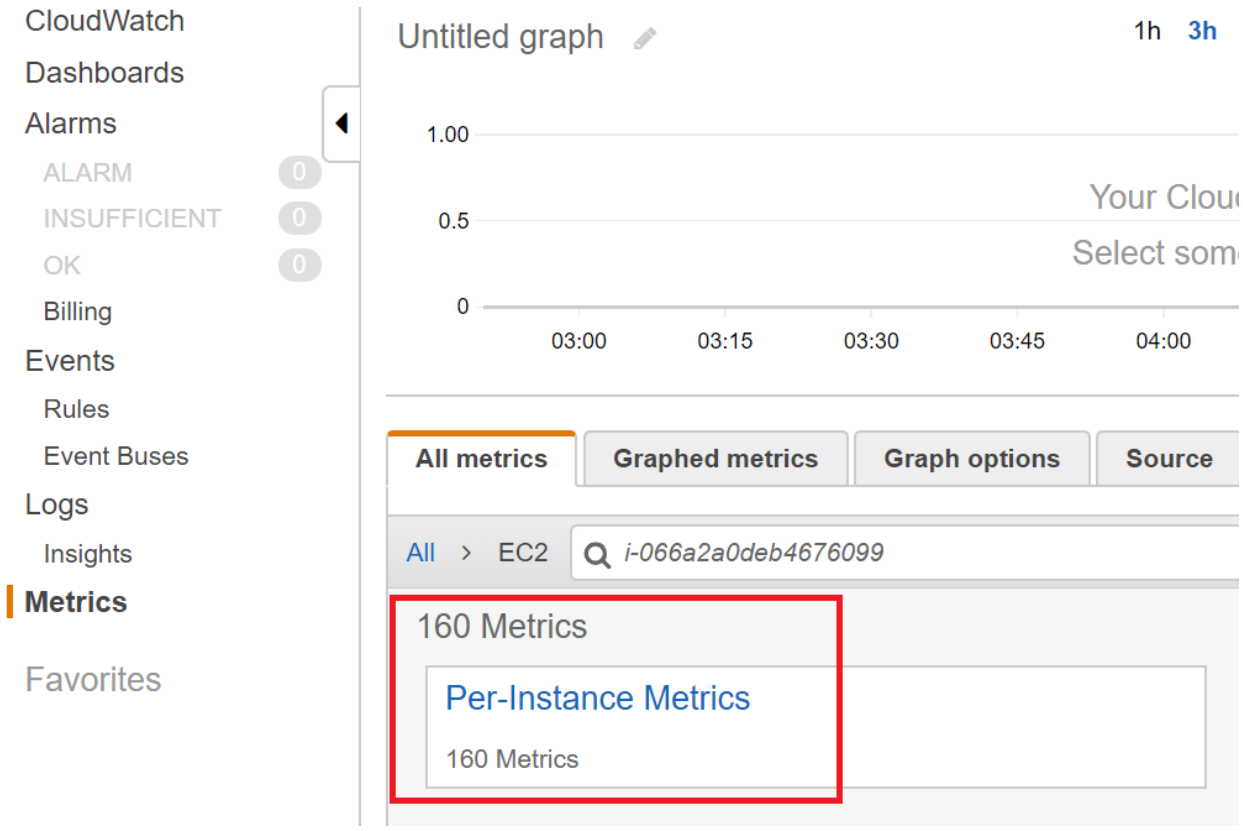
**Then click on Metrics**




**Copy the instance ID from Instance**

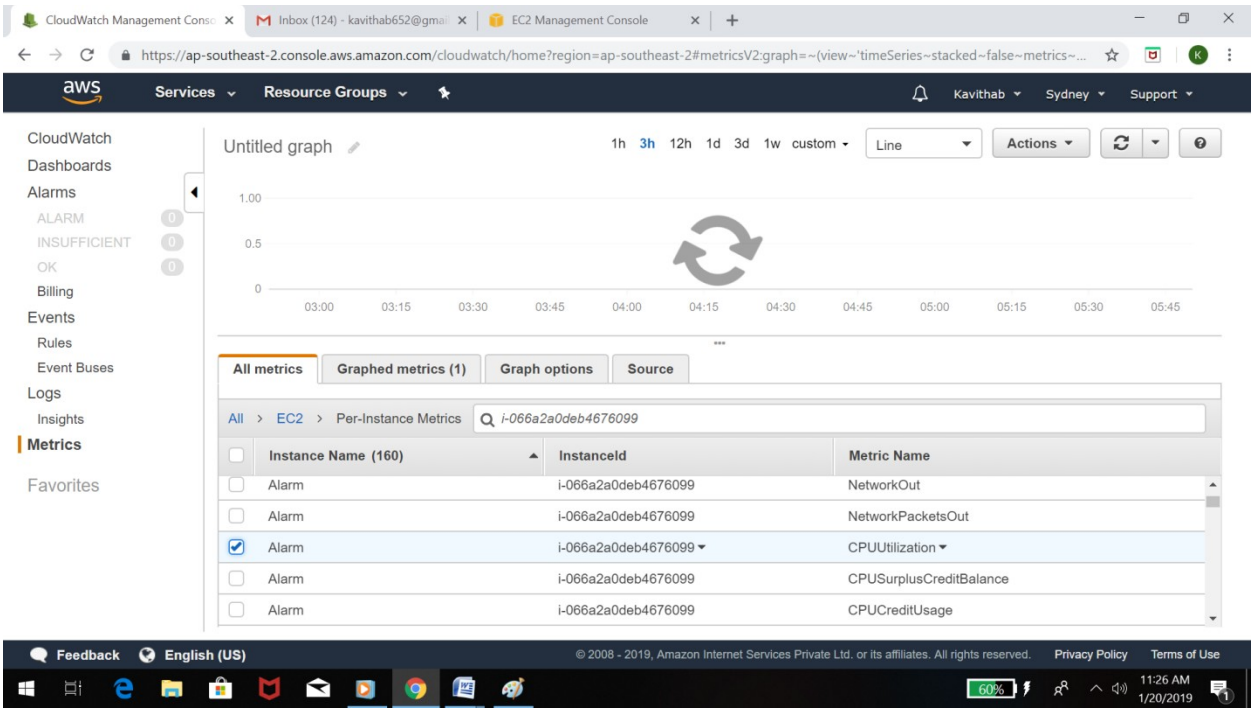


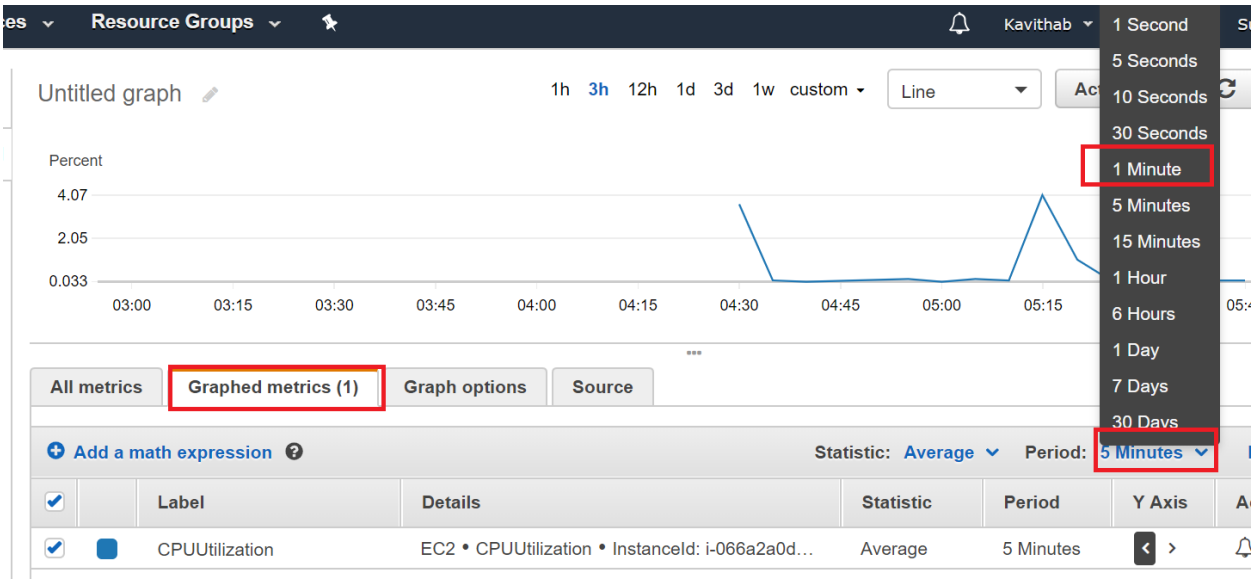

**Then go to Clouc watch -→Metrics -→ EC2---→ Pre instance Metrics**




**Then copy and paste the Instance id**

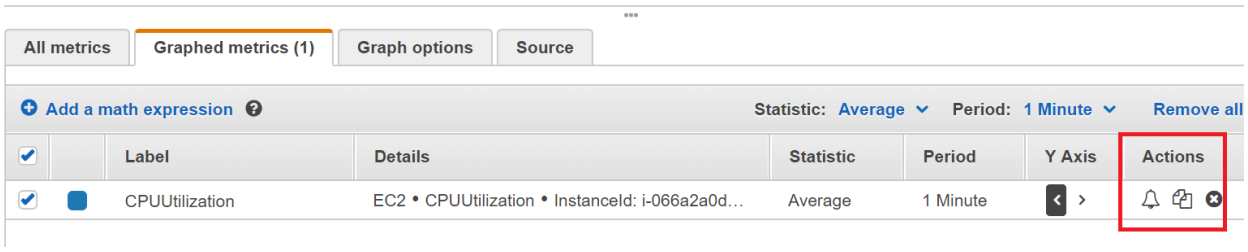
**Then select the Alarm of CPU Utilization from the given list below**



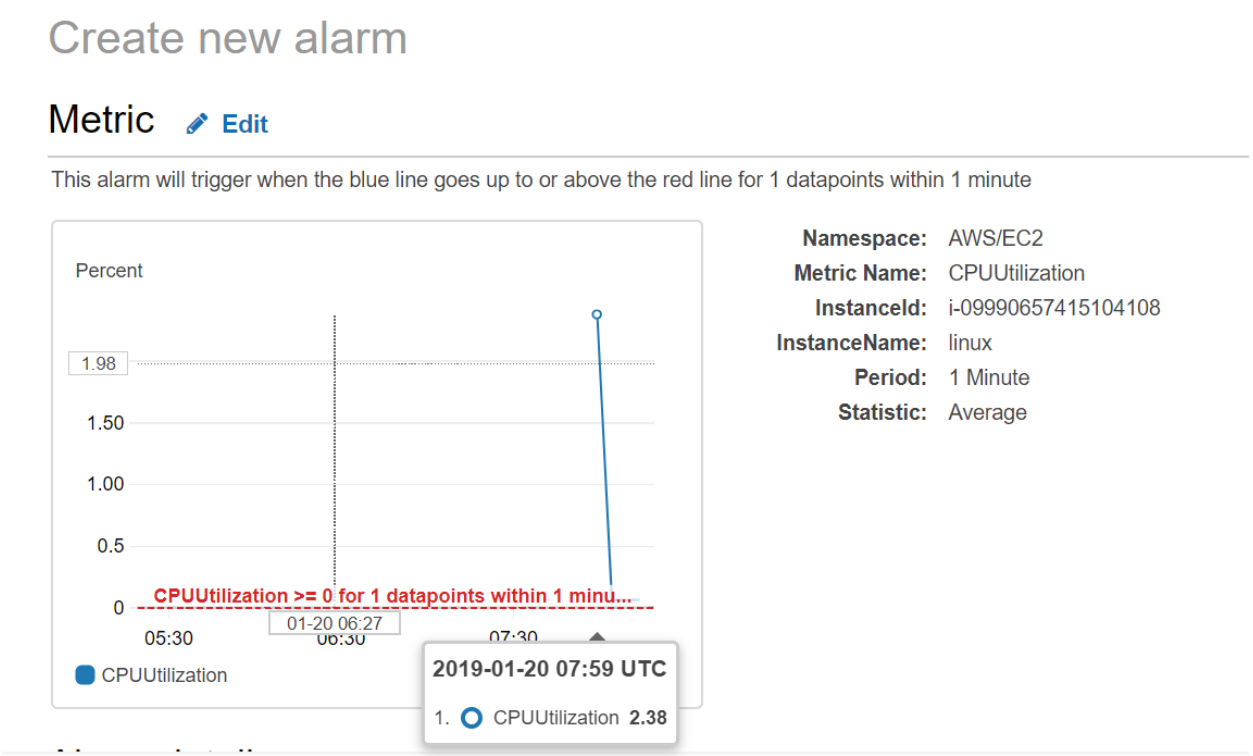




**Select Graphed Metrics -→ Drop down the Time Period - we can set the time as per our requirement**

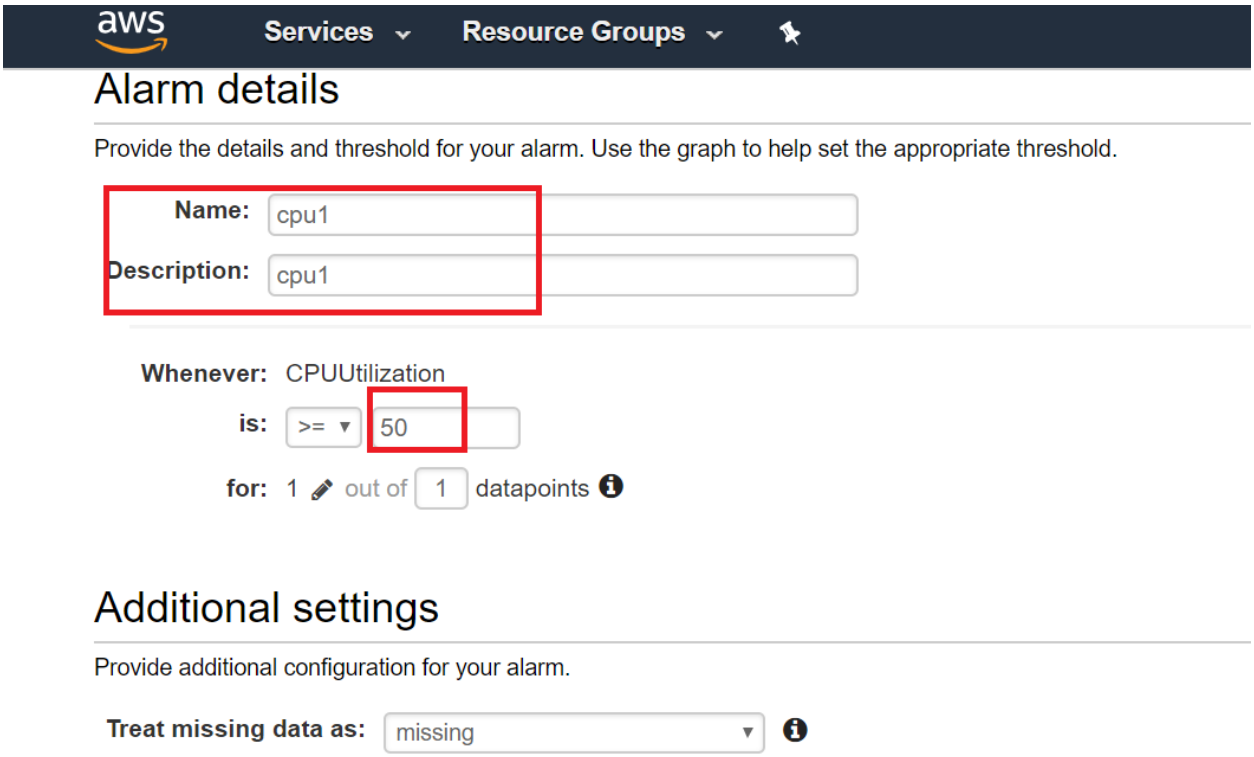
**In the above screen set for 1 minute**



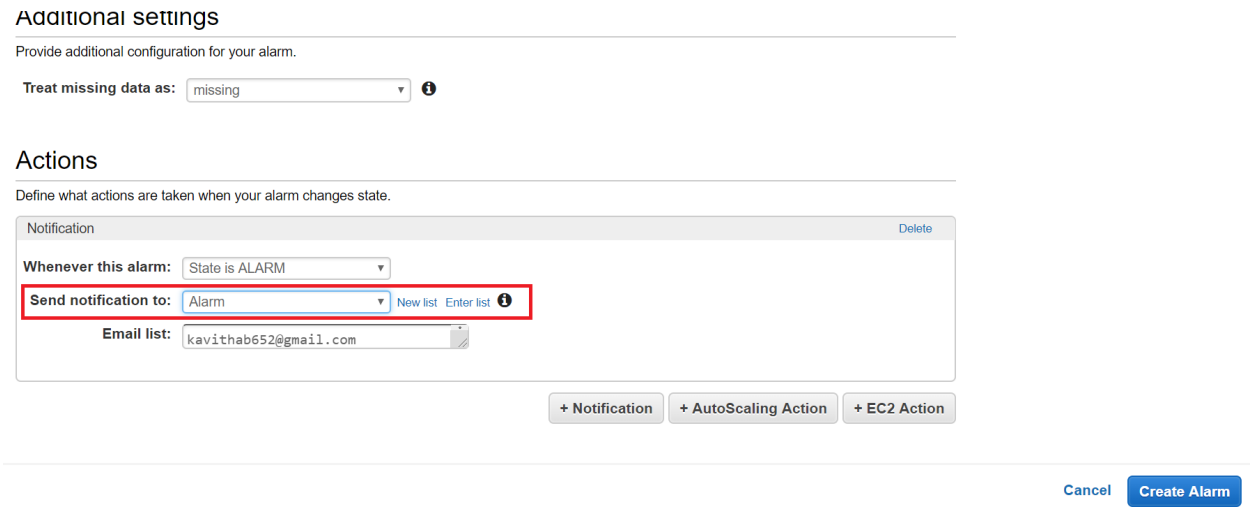

**Then Click on  Y Axis and then Click Bell Icon**



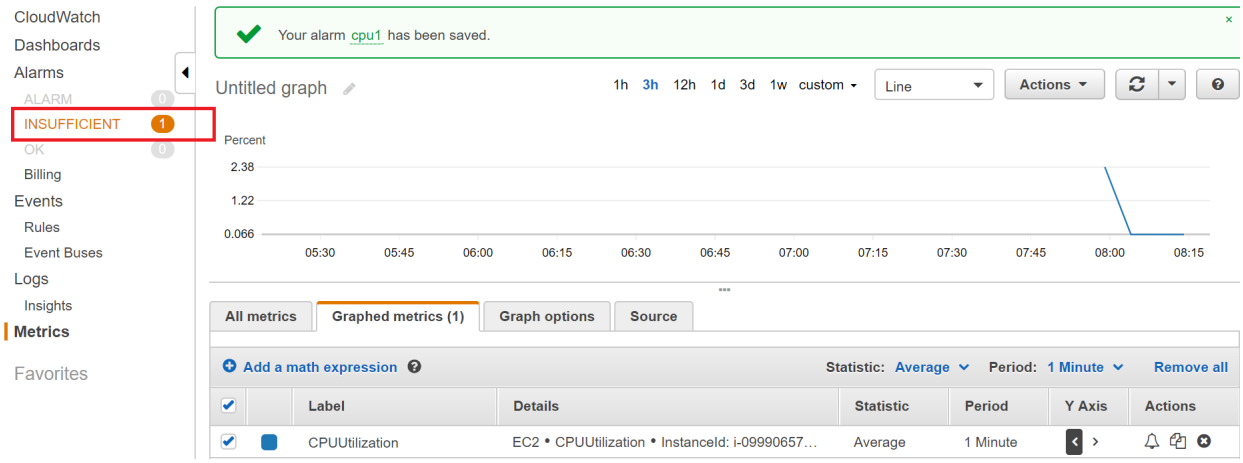

**Then Enter the name and description for Alarm as mentioned in the below screenshots**




**Select the Send notification to as Alarm from the drop down list**

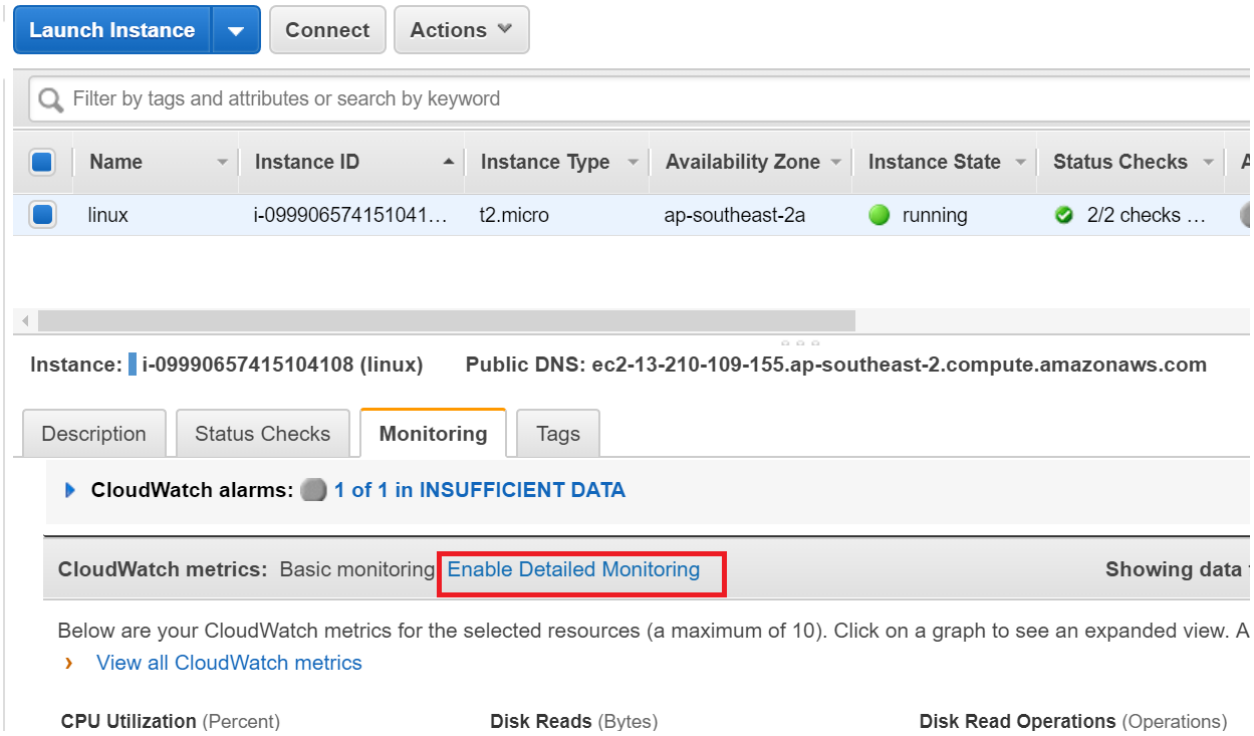


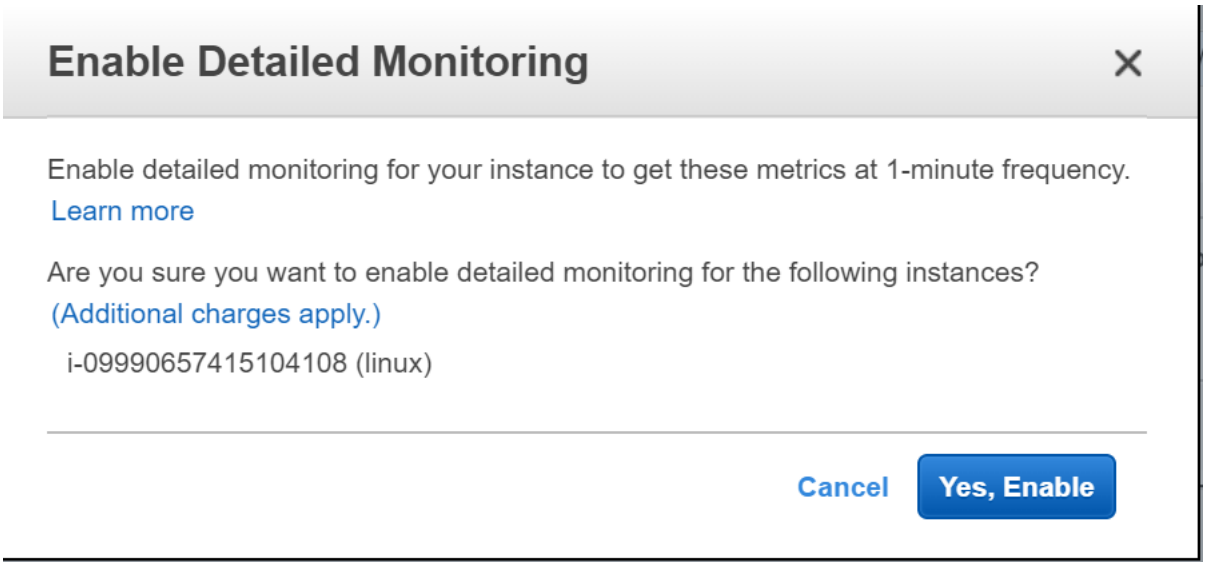

**Then Click on Create Alarm**




**Now we have to enable the detailed Monitoring**

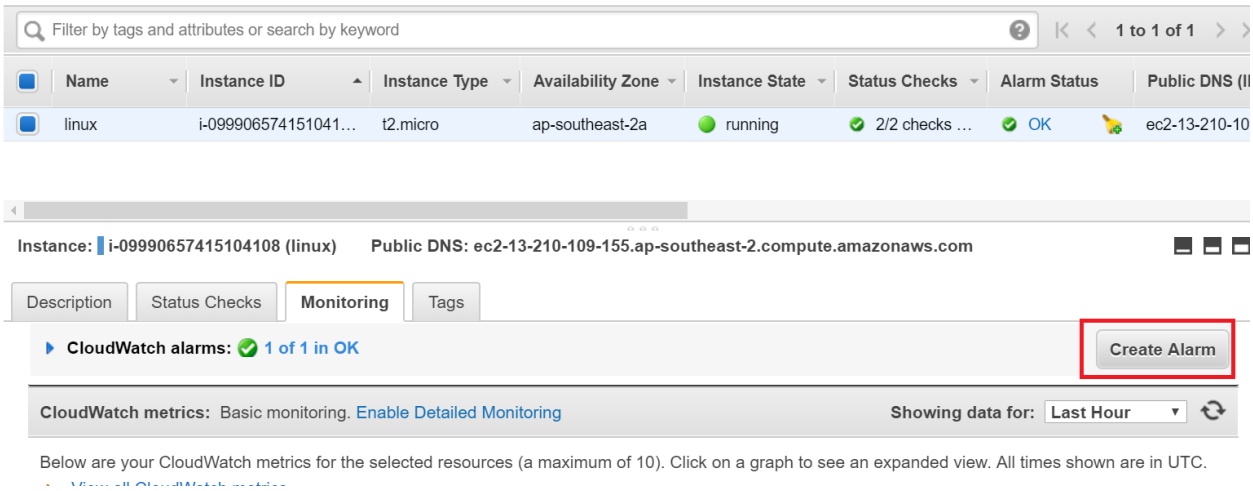
**Then go to Instance Right click on the Enable Detailed Monitoring to enable the**

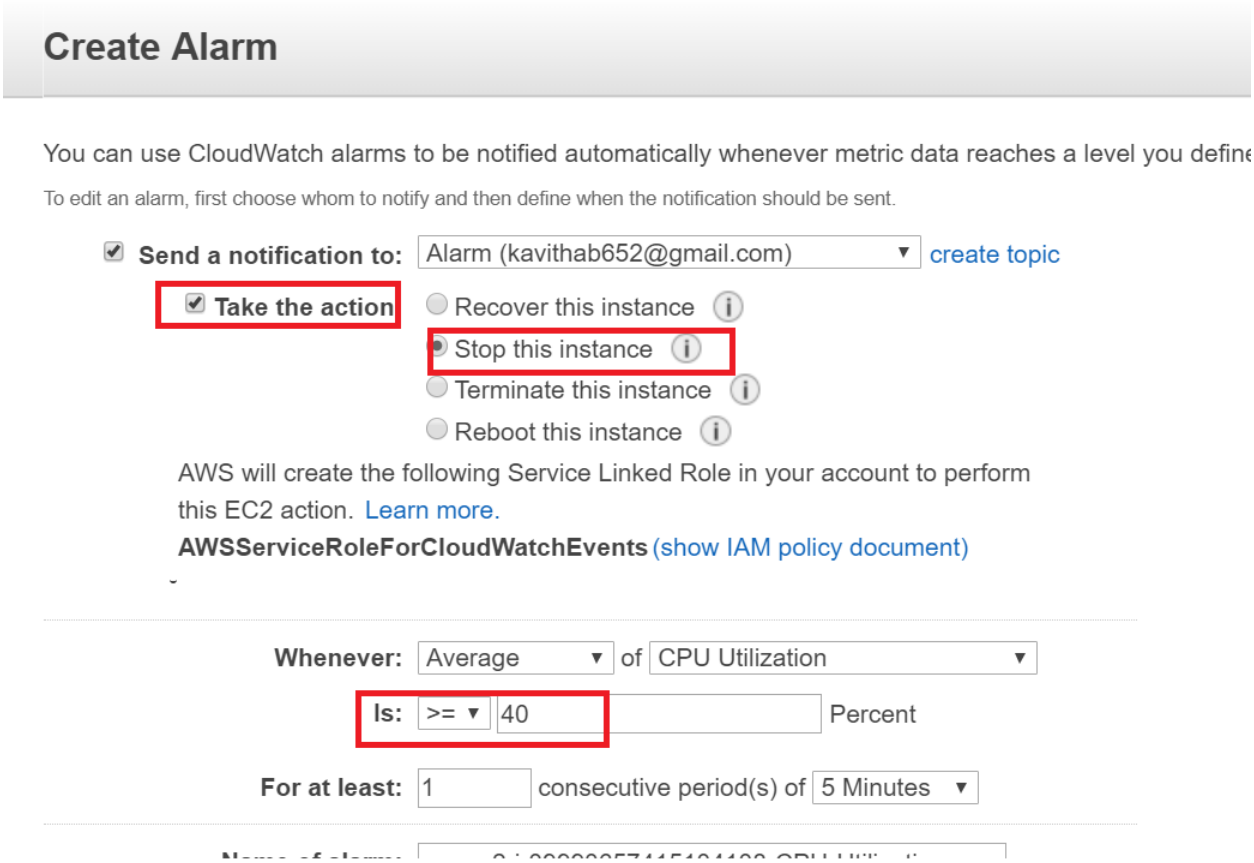





**Then Click on Yes Enable**

**Then go to Instance ---→ Monitoring ------click on Create Alarm**

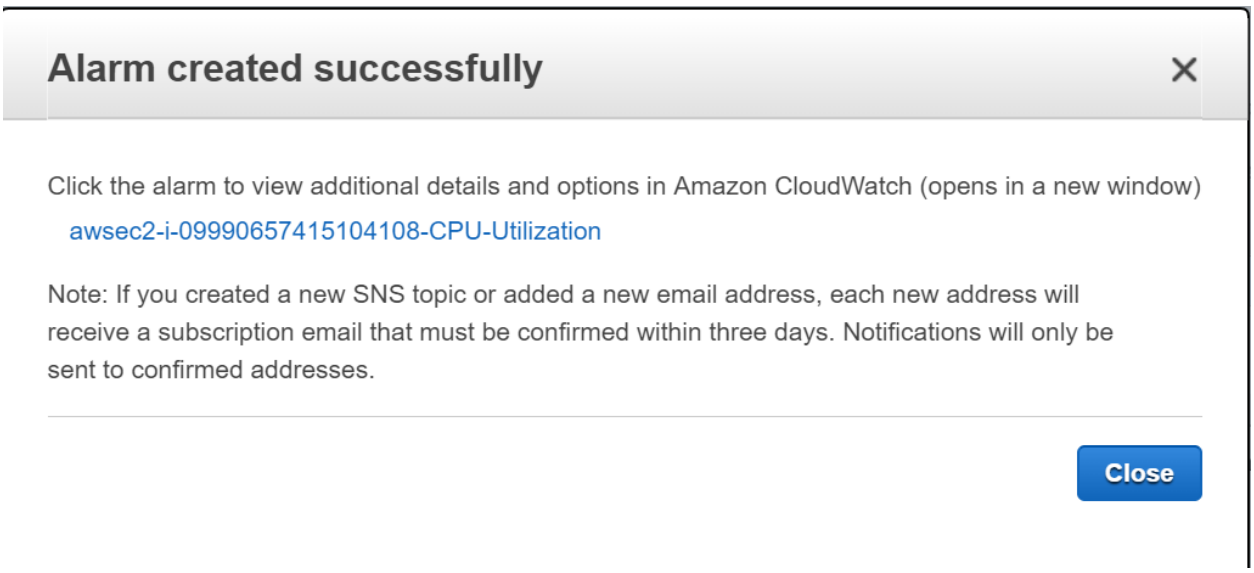





**Select the Take action Check box- Choose the option as per your requirement**

**Here we selected as Stop this instance**

**Set the threshold value**



**Go to OS level (Putty)**

**Login with User id - ec2-user**

**Execute below commands**

**# sudo –i**

**# top**

```
root@ip-10-0-0-4:~
top - 11:29:03 up  3:27,  1 user,  load average: 0.02, 0.01, 0.00
Tasks:  83 total,   1 running,  47 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
KiB Mem :  1009388 total,   617220 free,    59516 used,   332652 buff/cache
KiB Swap:        0 total,        0 free,        0 used.   790364 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S %CPU %MEM     TIME+ COMMAND
    1 root      20   0  125548   5404   4068 S  0.0  0.5   0:01.40 systemd
    2 root      20   0       0      0      0 S  0.0  0.0   0:00.00 kthreadd
    4 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 kworker/0:0H
    5 root      20   0       0      0      0 I  0.0  0.0   0:00.01 kworker/u30+
    6 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 mm_percpu_wq
    7 root      20   0       0      0      0 S  0.0  0.0   0:00.12 ksoftirqd/0
    8 root      20   0       0      0      0 I  0.0  0.0   0:00.15 rcu_sched
    9 root      20   0       0      0      0 I  0.0  0.0   0:00.00 rcu_bh
   10 root      rt   0       0      0      0 S  0.0  0.0   0:00.00 migration/0
   11 root      rt   0       0      0      0 S  0.0  0.0   0:00.03 watchdog/0
   12 root      20   0       0      0      0 S  0.0  0.0   0:00.00 cpuhp/0
   13 root      20   0       0      0      0 S  0.0  0.0   0:00.00 kdevtmpfs
   14 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 netns
   20 root      20   0       0      0      0 S  0.0  0.0   0:00.00 xenbus
   21 root      20   0       0      0      0 S  0.0  0.0   0:00.01 xenwatch
  170 root      20   0       0      0      0 S  0.0  0.0   0:00.00 khungtaskd
  171 root      20   0       0      0      0 S  0.0  0.0   0:00.00 oom_reaper
```

**In the above screen Utilization of CPU is 0%**

**To increase the cpu utilization execute the below command to generate alarm**

**# dd if=/dev/zero of=/dev/null**

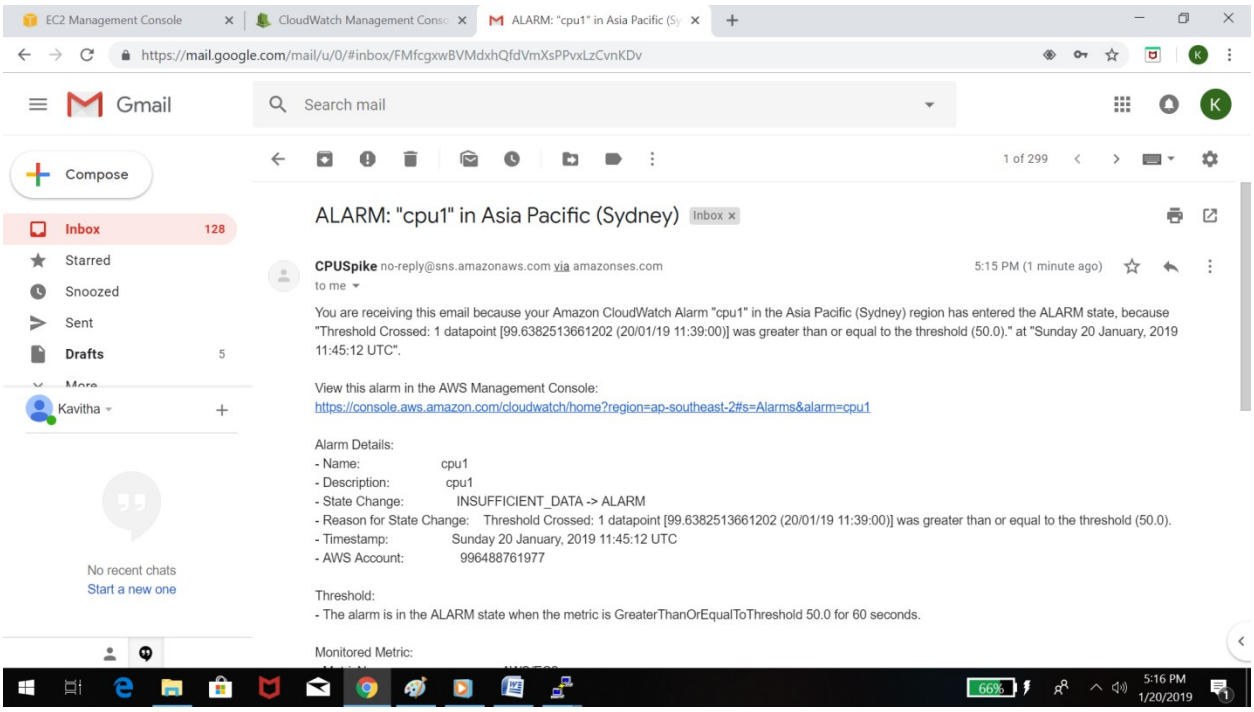
root@ip-10-0-0-4:~

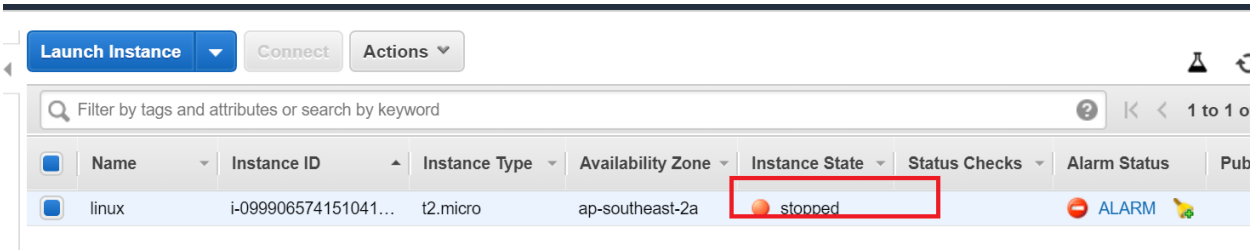
```
top - 11:38:25 up  3:36,  2 users,  load average: 0.15, 0.03, 0.01
Tasks:  89 total,   2 running,  52 sleeping,   0 stopped,   0 zombie
%Cpu(s): 25.0 us, 75.0 sy,  0.0 ni,  0.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
KiB Mem :  1009388 total,   610612 free,    65548 used,   333228 buff/cache
KiB Swap:        0 total,        0 free,        0 used.   783892 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S %CPU %MEM     TIME+ COMMAND
32648 root      20   0  114728    796    732 R 99.7  0.1   0:10.21 dd
    1 root      20   0  125548   5404   4068 S  0.0  0.5   0:01.41 systemd
    2 root      20   0       0      0      0 S  0.0  0.0   0:00.00 kthreadd
    4 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 kworker/0:0H
    5 root      20   0       0      0      0 I  0.0  0.0   0:00.01 kworker/u30:0
    6 root       0 -20       0      0      0 I  0.0  0.0   0:00.00 mm_percpu_wq
    7 root      20   0       0      0      0 S  0.0  0.0   0:00.12 ksoftirqd/0
    8 root      20   0       0      0      0 I  0.0  0.0   0:00.16 rcu_sched
    9 root      20   0       0      0      0 I  0.0  0.0   0:00.00 rcu_bh
```

**In the above screen CPU Utilization increased to 99.7%**

**We got the Alarm (alert) through mail**




**Once the Alarm generated and the instance is stopped.**